

Unlearn to Adapt: Using LLMs to Reveal Institutional Blind Spots in Climate Adaptation and Environmental Justice

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Climate adaptation is often framed as a process of learning: acquiring new knowledge, improving predictive models, and expanding institutional capacity. Yet many adaptation failures persist because institutions continue to rely on outdated assumptions, planning paradigms, and governance routines. We argue that climate adaptation also requires organizational unlearning: the deliberate questioning and retirement of practices that no longer support equitable, climate-resilient futures. We position organizational unlearning as an environmental and climate justice concern because frontline communities often experience the greatest consequences when institutions fail to abandon obsolete approaches. We further argue that recent advances in large language models (LLMs) create a new opportunity for Computer-Supported Cooperative Work (CSCW) research to examine institutional change at scale. Using post-Hurricane Katrina U.S. federal reports as a motivating case, we outline a research agenda combining AI-assisted discourse analysis with human interpretation to investigate how institutions reconsider, retain, or replace assumptions shaping climate adaptation and environmental justice.

Additional Key Words and Phrases: Organizational Unlearning; Climate Adaptation; Environmental Justice; Human-Centered AI

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1 INTRODUCTION

Climate change is increasingly characterized by compounding and cascading shocks, systemic disruptions, and growing uncertainty, placing unprecedented pressure on institutions responsible for climate adaptation [15, 19, 29]. Because institutions decide how to respond to that pressure, climate adaptation is therefore not only a technical challenge but also a question of environmental and climate justice: adaptation decisions determine whose knowledge informs policy, whose experiences are recognized, and which communities bear the greatest consequences when institutions fail to respond to changing environmental conditions. While adaptation research and practice have largely emphasized learning through scientific innovation, improved forecasting, and institutional capacity building [5, 9, 32], this dominant framing overlooks an equally important challenge. Adaptation failures are often rooted in institutional adherence to legacy assumptions, planning paradigms, and governance routines that no longer reflect contemporary climate realities.

Despite substantial investments in climate science and resilience planning, vulnerability remains entrenched, billion-dollar disasters are becoming more frequent, and adaptation efforts can inadvertently reinforce existing inequalities or produce maladaptation [2, 10, 16, 17]. This pattern reflects a broader challenge: institutions often possess substantial technical expertise yet continue to rely on assumptions, standards, and decision-making processes developed for historical

climate conditions rather than today’s increasingly uncertain risk landscape. We argue that climate adaptation requires not only learning but also organizational unlearning [8, 28]: *the deliberate process of questioning, retiring, and replacing assumptions, routines, and governance practices that no longer support public safety, resilience, or equitable adaptation*. Organizational unlearning is not only an internal organizational process but also an environmental and climate justice concern, as outdated institutional logics shape whose knowledge is valued and whose experiences remain marginalized. Although rarely discussed explicitly in climate adaptation scholarship, its importance is reflected in calls for transformative adaptation, the rethinking of entrenched assumptions, and the recognition of Indigenous and community knowledge [13, 18, 22].

Studying organizational unlearning, however, remains difficult because evidence is distributed across thousands of pages of government reports, engineering manuals, congressional testimony, technical guidance, and policy documents. Recent advances in Large Language Models (LLMs) provide a promising opportunity to augment this work by supporting the analysis of large and longitudinal policy corpora [1, 20, 27]. Rather than treating LLMs merely as tools for document classification or summarization, we argue that they can assist researchers in identifying where institutions acknowledge past failures, challenge legacy assumptions, retire obsolete practices, and integrate alternative forms of knowledge. Importantly, we envision these systems as supporting, rather than replacing, human interpretation and collaborative sensemaking, aligning with CSCW’s longstanding emphasis on human-centered technologies.

As a motivating example, we consider the institutional reforms that followed *Hurricane Katrina*. The post-Katrina transformation of the U.S. Army Corps of Engineers illustrates how longstanding engineering assumptions and organizational practices shifted from a paradigm of absolute protection toward one focused on risk reduction and resilience [11]. Driven by both engineering failures and the disproportionate impacts on frontline communities, this case demonstrates how institutions can rethink outdated planning assumptions. Rather than presenting a completed empirical study, we use it to illustrate how CSCW can investigate institutional change through computational approaches grounded in human expertise and environmental justice. Through this provocation, we argue that organizational unlearning is an overlooked yet essential lens for CSCW research on environmental and climate justice. We further propose that AI-assisted analysis of institutional discourse can help researchers understand not only what organizations learn, but also the assumptions they are willing—or unwilling—to let go of, creating new opportunities for transparent, accountable, and justice-oriented climate governance.

1.1 Organizational Unlearning as an Environmental Justice Lens

Organizational unlearning has been studied across management, organizational behavior, and psychology as the intentional process of questioning and abandoning obsolete assumptions, routines, and knowledge systems that constrain adaptation and innovation [4, 7, 14, 21]. While these perspectives primarily focus on organizational effectiveness, unlearning is equally relevant to environmental and climate justice. Decisions about what institutions retain, revise, or abandon influence which knowledge is considered legitimate, which experiences shape policy, and whether frontline communities continue to be marginalized in climate decision-making. We view organizational unlearning through four complementary dimensions. Institutions may *i) reconsider* dominant knowledge systems by recognizing community and Indigenous expertise; *ii) discard* ineffective policies and governance routines; *iii) realign* technical standards in response to changing climate risks; and *iv) merge* previously disconnected domains such as climate adaptation, resilience, and environmental justice. Together, these dimensions provide a practical vocabulary for examining institutional change while encouraging CSCW researchers to investigate not only how technologies support organizational learning, but also how they help reveal outdated assumptions and create space for more equitable forms of knowledge and participation.

2 LLMS AS A METHODOLOGICAL PROVOCATION FOR CSCW

Existing computational approaches have been successful at identifying topics, sentiment, and emerging themes in policy text. Still, they rarely examine whether institutions reconsider or abandon the assumptions that shape decision-making. Rather than using LLMs to summarize documents, we propose using them to identify discourse associated with organizational unlearning. Government reports, engineering manuals, congressional testimony, and policy evaluations frequently contain signals of organizational change that are difficult to identify manually due to their distribution across thousands of pages. LLM-assisted analysis can help surface passages in which agencies acknowledge prior failures, revise planning assumptions, retire obsolete practices, or incorporate new forms of expertise. At the same time, LLMs carry their own biases as instruments of social measurement: they can privilege dominant institutional voices and vocabularies, misread ambiguous or bureaucratic language as more decisive than it is, and impose contemporary framings on historical text. We therefore treat the provocation itself as an object of evaluation, not just a method to apply, and propose systematically assessing where LLM-identified instances of unlearning align with, diverge from, or miss what domain experts identify through close reading. Human interpretation remains essential for validating findings and situating computational results within their organizational and political contexts. The full codebook and LLM prompt used for this analysis are provided in Appendix A and Appendix B for reference and reproducibility.

Hurricane Katrina provides a compelling, motivating case. The disaster exposed profound institutional failures [31] while disproportionately affecting marginalized communities [6]. Many post-Katrina reforms focused on improving infrastructure and operational capacity [30], yet a key question remains: **which assumptions were genuinely abandoned, and which continued to shape adaptation planning?** Two decades later, community-led work in the same region continues to grapple with this question directly, designing resilient communication and mutual aid infrastructure precisely because institutional systems remain unreliable [12]. We argue that answering this question requires computational methods capable of examining institutional discourse beyond conventional thematic analysis.

3 TOWARD A CSCW RESEARCH AGENDA

We believe CSCW is uniquely positioned to advance this agenda by developing and critically evaluating computational methods that make institutional adaptation more transparent, accountable, and open to scrutiny. Building on the CSCW and HCI communities’ longstanding work on disaster response, institutional data, and sustainability [3, 23–26], organizational unlearning complements existing approaches by focusing on the retirement of obsolete assumptions, governance routines, and technical standards. Combined with LLMs, this perspective enables researchers to examine not only what institutions learn, but also what they leave behind—while critically assessing whether AI systems reproduce the same institutional blind spots they aim to uncover. We encourage the CSCW community to pursue three complementary directions: (1) *develop and evaluate computational methods for identifying institutional unlearning across large policy archives, including their limitations and biases*; (2) *design transparent human-AI workflows that keep domain experts and affected communities central to interpretation*; and (3) *investigate how these methods can support more inclusive and equitable climate governance*. Ultimately, climate adaptation should be evaluated not only by the knowledge it acquires, but also by its willingness to question inherited assumptions and retire outdated practices. Used critically, LLMs can help reveal where such institutional change occurs—and where it does not. We hope this perspective encourages CSCW researchers to view AI not simply as a document-analysis tool but as a methodological partner for studying institutional change, environmental justice, and accountable climate governance.

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APPENDIX

A ORGANIZATIONAL UNLEARNING CODEBOOK

Figure 1 presents the organizational unlearning codebook used for paragraph-level annotation of government policy documents. The organizational unlearning codebook was iteratively developed and refined through multiple rounds of discussion involving three graduate student researchers and four domain experts.

B LARGE LANGUAGE MODEL PROMPT

The following prompt template was used across all LLMs in this study. The system instruction, task description, codebook, decision rules, and output schema remained constant across models. Only the paragraph identifier and paragraph text changed for each annotation task.

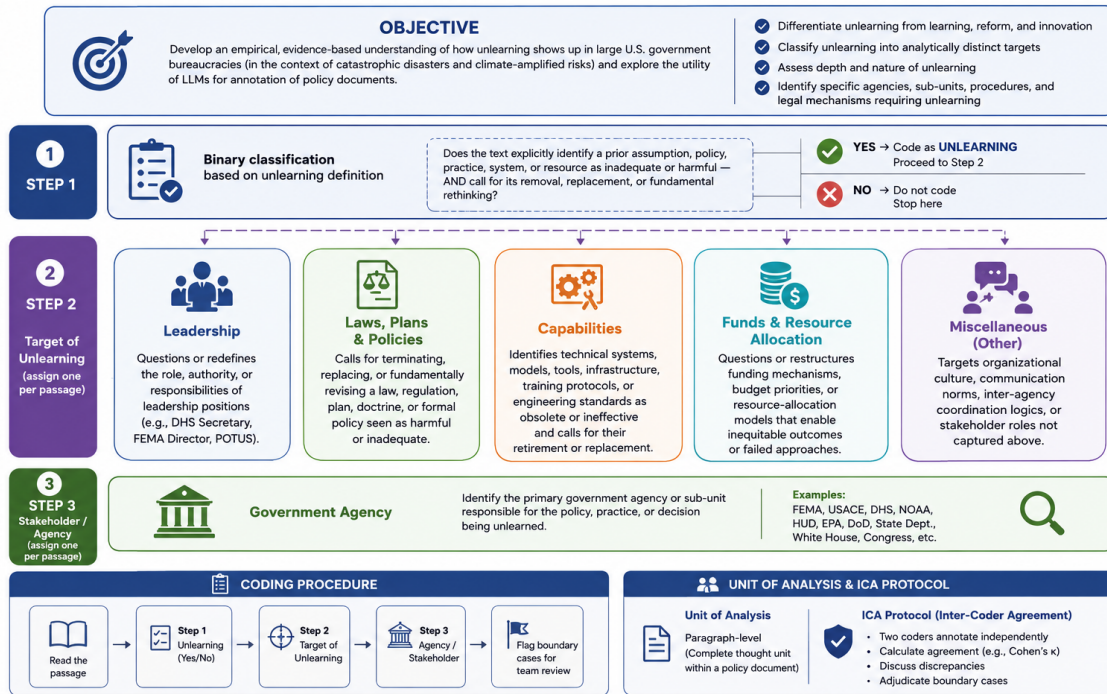


Figure X. Revised codebook for annotating unlearning in government policy documents.

The codebook operationalizes unlearning as a binary classification, five target categories, and agency identification at paragraph level.

Fig. 1. Codebook used for paragraph-level annotation. The annotation workflow consists of three sequential stages: (1) determining whether a passage contains evidence of organizational unlearning, (2) assigning a single target of unlearning (leadership, laws/plans/policies, capabilities, funds and resource allocation, or miscellaneous organizational practices), and (3) identifying the relevant government agency or institutional actor. Boundary cases were flagged for team adjudication, and inter-coder agreement was assessed on paragraph-level annotations.

B.1 System Prompt

System Prompt

Return valid JSON only. Do not include Markdown, code fences, commentary, or any text outside the JSON object.

B.2 Example User Prompt

Example User Prompt

You are coding federal disaster policy text using the organizational unlearning codebook.

Task

For the paragraph provided at the end of this prompt:

- (1) Decide whether the paragraph contains evidence of organizational unlearning.
- (2) If unlearning is present, assign exactly one target of unlearning.
- (3) Identify the government agency, office, sub-unit, leadership body, or institutional actor whose prior assumption or practice is being challenged, but only when it is explicitly stated.
- (4) Provide a confidence score between 0.0 and 1.0.
- (5) Give a short rationale of one sentence.

Important Coding Principles

- Code at the **paragraph level**. Treat the entire paragraph as the unit of analysis.
- Do not classify a passage as unlearning based on tone, keywords, criticism, or problem diagnosis alone.
- A passage must identify a prior assumption, policy, practice, routine, structure, or technical system as inadequate, obsolete, harmful, or misaligned **and** call for its removal, replacement, abandonment, or fundamental rethinking.
- Purely additive change is not unlearning. Adding resources, personnel, technology, training, coordination, or capacity without rejecting the prior underlying logic should be coded as no unlearning.
- Routine reform, incremental improvement, modernization, or better implementation is not necessarily unlearning.
- If the evidence is ambiguous, return no unlearning.
- If unlearning is not present, set the target to none.
- Assign exactly one target type when unlearning is present.

Codebook

Step 1: Unlearning (Yes/No)

Organizational unlearning is the deliberate process by which a government agency discards misaligned assumptions, obsolete knowledge, or outdated routines in response to demonstrated failure. Unlearning becomes visible in discourse when an institution explicitly problematizes a prior practice, justifies its abandonment, and signals a departure from established norms.

The three central attributes are:

- **Intentionality:** The departure is deliberate and purposeful rather than passive forgetting.

- **Subtraction:** The passage removes, abandons, or rejects something previously accepted rather than merely adding something new.
- **Discontinuity:** The change represents a meaningful departure from inherited assumptions, objectives, norms, or approaches rather than a minor adjustment.

Detection question:

Does the paragraph explicitly identify a prior policy, assumption, practice, routine, governance structure, or technical system as inadequate or harmful **and** call for its removal, replacement, abandonment, or fundamental rethinking?

- If yes, code `unlearning_present` as true and proceed to Step 2.
- If no, code `unlearning_present` as false, set `target_type` to none, and stop the classification process.

Positive indicators

Code as unlearning when the paragraph:

- names a specific prior assumption, policy, practice, routine, system, or governance arrangement as a source of failure;
- explicitly calls for abandoning, dismantling, replacing, retiring, or fundamentally rethinking it; and
- frames the departure from prior practice as necessary rather than merely desirable.

Negative indicators

Do not code as unlearning when:

- the paragraph only documents what went wrong without calling for change;
- the paragraph recommends adding resources, personnel, technology, training, or capacity without rejecting the prior logic;
- the paragraph uses “lessons learned” language that implies improvement rather than replacement;
- a new tool or system is adopted while the underlying institutional assumption remains unchanged;
- the passage describes routine maintenance, modernization, coordination, or policy implementation; or
- the passage is aspirational but does not identify a prior practice that should be abandoned.

Boundary rule

Problem diagnosis alone is not unlearning. For example, stating that a command structure was inadequate is not sufficient unless the paragraph also calls for dismantling, replacing, or fundamentally redesigning that structure.

Step 2: Target of Unlearning

If unlearning is present, assign exactly one of the following targets.

A. Leadership

Use Leadership when the paragraph questions or redefines the role, authority, accountability, or responsibilities of a leadership position or leadership structure.

Code this target when the paragraph:

- questions whether an existing leadership structure is fit for catastrophic disaster management;
- calls for consolidating, redistributing, or redefining leadership authority;
- rejects a distributed or unclear authority structure; or

- frames prior leadership arrangements as a source of institutional failure.

Do not use this target when:

- the paragraph merely recommends appointing a new person to an unchanged role;
- it proposes a minor reporting-line adjustment;
- leadership is mentioned only as context; or
- the primary object of change is a law, technical system, funding structure, or coordination practice.

B. Laws, Plans, and Policies

Use `laws_plans_policies` when the paragraph calls for terminating, replacing, removing, or fundamentally revising a law, regulation, executive order, plan, doctrine, or formal policy.

Code this target when the paragraph:

- identifies a specific legal or policy arrangement as obsolete, harmful, unjust, or structurally inadequate;
- calls for eliminating or replacing a statute, regulation, doctrine, plan, or policy;
- frames the formal policy arrangement as a root cause of failure; or
- proposes dismantling an existing governance framework rather than modifying it at the margins.

Do not use this target when:

- the passage recommends better implementation of an unchanged policy;
- a new plan is added alongside the existing plan without rejecting it;
- the change is a routine policy update; or
- the passage calls only for clarification, enforcement, or minor amendment.

C. Capabilities

Use `capabilities` when the paragraph identifies a technical or operational system as based on obsolete assumptions and calls for its recalibration, replacement, or fundamental redesign.

Capabilities may include:

- risk models;
- maps;
- software;
- communications systems;
- infrastructure;
- engineering standards;
- damage-assessment systems;
- citizen-service systems;
- training protocols;
- disaster exercises; or
- operational procedures.

Code this target when the paragraph:

- identifies a technical system or protocol as inadequate for catastrophic or climate-amplified risk;
- calls for replacing an engineering standard or risk model;

- calls for overhauling an existing training, exercise, communication, assessment, or operational regime; or
- links technical failure to an obsolete underlying assumption.

Do not use this target when:

- the passage describes routine maintenance;
- equipment is upgraded without questioning the existing technical logic;
- new technology is added only as a supplement; or
- the passage identifies a capability gap but does not call for replacing the system that produced it.

D. Funds and Resource Allocation

Use `funds_resources` when the paragraph questions or changes the rules, authorities, priorities, or structures governing how disaster resources are allocated, distributed, accepted, or managed.

Code this target when the paragraph:

- calls for replacing an existing funding or resource-allocation mechanism;
- identifies current allocation rules as a source of failure or inequity;
- calls for changing the authority governing resource distribution;
- rejects an existing budgeting or resource-management logic; or
- proposes fundamentally restructuring how financial, material, or external assistance is managed.

Do not use this target when:

- the paragraph only requests more money, personnel, or supplies;
- a new grant program is added without rejecting an existing allocation logic;
- funding levels change while the underlying allocation process remains the same; or
- resources are mentioned only as context.

E. Miscellaneous Organizational

Use `misc_organizational` only when the paragraph explicitly rejects a prior organizational assumption or practice that cannot reasonably be assigned to the four primary targets above.

Possible domains include:

- organizational culture;
- communication norms;
- inter-agency coordination logic;
- stakeholder participation;
- public-involvement practices;
- information-sharing norms;
- professional identity; or
- other organizational routines not captured above.

Use this category sparingly.

Do not use it:

- as a default category for ambiguous passages;

- when the paragraph can reasonably be assigned to Leadership, Laws/Plans/Policies, Capabilities, or Funds/Resources;
- when the passage only recommends improved coordination or communication without rejecting a prior practice; or
- when the passage contains general organizational change but no evidence of unlearning.

Step 3: Government Agency or Institutional Actor

Record the government agency, office, sub-unit, leadership body, or institutional actor whose prior practice, assumption, policy, or system is the object of unlearning.

- Code the agency at the most specific level explicitly stated in the paragraph.
- Code the agency whose practice must change, not the agency issuing the recommendation.
- For example, if GAO recommends that FEMA abandon a prior practice, record FEMA rather than GAO.
- Do not infer an agency from the source document, metadata, title, or outside knowledge.
- If no agency or sub-unit is explicitly stated in the paragraph, return null.

Output Consistency Rules

- If unlearning_present is false, then target_type must be none.
- If unlearning_present is true, then target_type must be exactly one of the five target categories.
- Do not return multiple targets.
- The agency may be null whether unlearning is present or absent.
- The confidence value must be a number from 0.0 to 1.0.
- The rationale must be one short sentence explaining the decisive evidence.
- Do not include additional fields.

Return only valid JSON using the following structure:

```
{
  "unlearning_present": true,
  "target_type": "Leadership",
  "agency": "Federal Emergency Management Agency",
  "confidence": 0.92,
  "rationale": "The paragraph rejects the prior distributed leadership structure and calls for unified authority."
}
```

The allowed values for target_type are:

```
"Leadership"
"laws_plans_policies"
"capabilities"
"funds_resources"
"misc_organizational"
"none"
```

Paragraph ID

<Insert paragraph identifier here>

Paragraph to Annotate

<Insert paragraph text here>